

Research Project (IT4010) – Proposal Presentation Justification Sheet

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Project ID: 25-26J-427

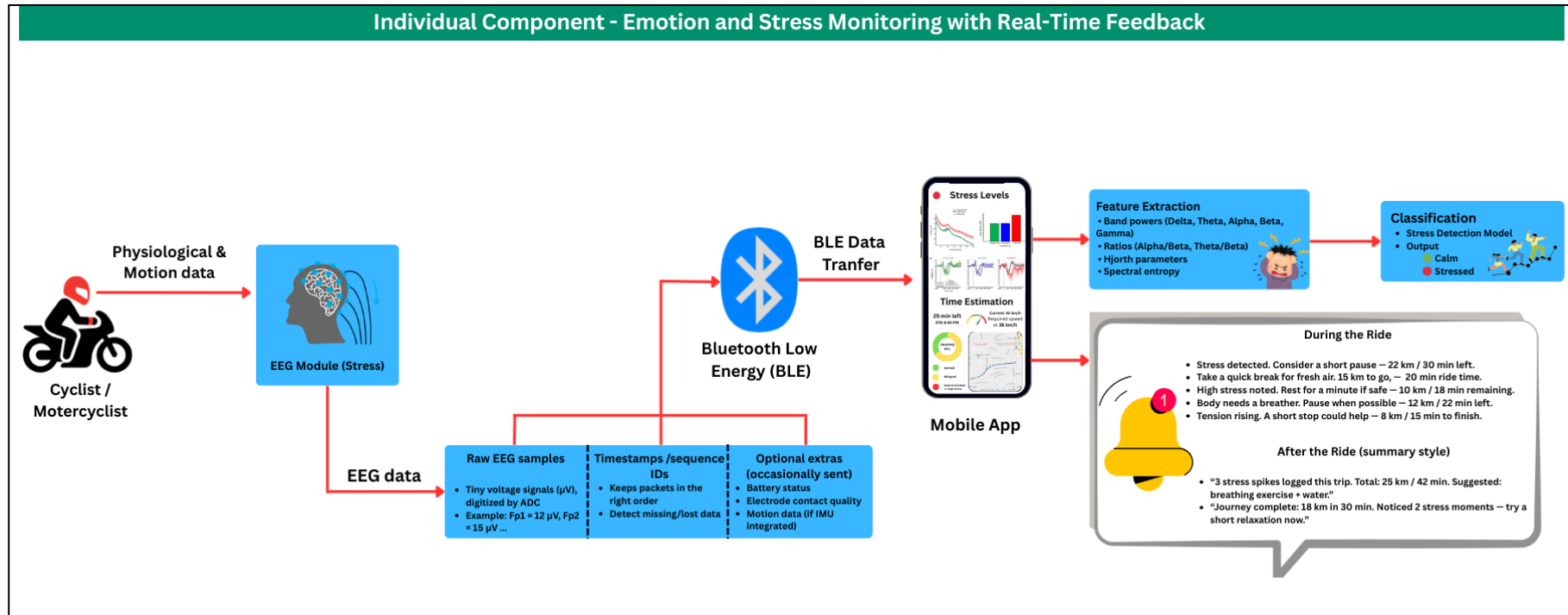
Title of the Project: Smart Helmet for Cyclists and Motor Cyclists

Individual Component in Brief (Single Sentence): Emotion and Stress Monitoring with Real-Time Feedback

Sub Assessment Criteria	Excellent (100 - 75)	Good (74 - 60)	Average (59 - 40)	Below Avg. (39 - 0)	Justification
Proven gap/Creative Solution [Based on LO1] - [35%]					
Knowledge gap (Problem) with novel and creative solution 70%	Clearly argued the existence of knowledge gap using credible, multiple sources. Excellent justification for novelty and creativity.	Sufficiently argued the existence of knowledge gap using limited sources. Good justification for novelty and creativity.	Moderately argued the knowledge gap with very few or no sources. Average justification for novelty and creativity.	Knowledge gap is not clearly identified. Poor or No justification for novelty and creativity.	Most helmets protect only physically, while wearables track heart rate but not cognitive stress. Our system uses EEG with IoT and mobile feedback for real-time stress detection, personalized relaxation, and alerts, making helmets emotionally active safety companions.
Compare existing systems and related work 30%	Extensively compared the research problem in the present context of the research domain / with similar products and services.	Sufficiently compared the research problem in the present context of the research domain / with similar products and services.	Moderately compared the research problem in the present context of the research domain / with similar products and services	Poorly compared or No comparison has been done	Existing intelligent helmets focus on crash warnings and physical well-being, while wearables estimate strain based on heart variability, an insecure measure when on a ride. We use EEG directly with BLE, classification, and real-time indication, allowing for more accurate, responsive monitoring of emotions than available solutions.
Capability in applying the knowledge in particular stream [Based on LO2] - [30%]					
Application of key pillars in the specialized area of knowledge 50%	Clearly identified the most appropriate research area (Machine learning, Image processing, Data Science etc.)	Satisfactorily identified the research area.	Vaguely identified. But appropriateness is doubtful.	Indistinctly identified or Not identified.	The module applies biomedical signal processing, BCI, BLE communication, and machine learning on EEG data to deliver seamless, real-time stress detection and responsive feedback during rides.
Application of technologies in the relevant key pillar/area 50%	Demonstrated full awareness of technologies in the relevant area and a critical evaluation of technologies proving selection of the best technology/s.	Adequate awareness of technologies to be used and a good evaluation of technologies proving selection of the best technologies.	Moderate awareness of technologies to be used.	Poor or Zero awareness of technologies to be used.	The system employs EEG sensing, ESP32 with BLE, and a mobile app for real-time stress detection. TensorFlow Lite classifies stress peaks, delivering instant alerts with relaxation guidance, ensuring scalability, portability, and effective edge analytics.
Solution Implementation [Based on LO3] - [5%]					

High-level System Architecture and identification of selfevaluation plan/criteria 50%	Brilliantly justified high-level SA with highly acceptable self-evaluation plan.	Adequately justified high-level SA with acceptable self-evaluation plan.	Barely justified high-level SA with fairly acceptable selfevaluation plan.	Poorly justified or No evidence of High-level SA with inappropriate or No selfevaluation plan.	It integrates an EEG module, BLE-capable ESP32, and a mobile app for stress classification, visualizations, and notification. Evaluation considers model performance, low-latency notification, usability feedback from the rider, and stress reduction through guided interventions. It includes Appendix 1.
User Requirements / Functional Requirements 20%	Comprehensive and realistic user requirements and the functional requirements well described.	Comprehensive and realistic user requirements and the functional requirements adequately described.	Comprehensive and realistic user requirements and the functional requirements barely described.	Comprehensive and realistic user requirements and the functional requirements poorly described.	It records and transmits EEG data in real time, determines stress spikes, determines the emotional state, provides immediate alert with a request for relaxation, displays real-time status on the mobile app, and allows customizable notification options.
Work Breakdown Structure (WBS) 30%	Comprehensive planning demonstrate in WBS, realistic time estimates and right workload distribution.	Good planning demonstrate in WBS, realistic time estimates and good workload distribution.	Satisfactory planning of WBS, barely feasible time estimates and acceptable workload distribution.	Poor or No planning of WBS, Unrealistic time estimates and unacceptable workload distribution.	The work plan spans 18 weeks: initial literature review and sensor selection, EEG integration with BLE, stress model training, mobile app development, system integration and testing, followed by user trials, evaluation, and final refinements. Included Appendix 2
Effective Communication [Based on LO4] - [15%]					Will be assessed entirely at the Presentation
Ability of commercialization / potential for entrepreneurship [Based on LO5] - [15%]					
Ability of commercialization / potential for entrepreneurship 100%	Demonstrated sound evidence to prove business potential highlighting many achievable user benefits	Sufficient evidence to prove business potential highlighting some achievable user benefits	Few evidence to prove business potential with few user benefits	Very few or No evidence to prove business potential with unachievable or No user benefits	It has great commercialization potential as a first-of-its-kind mental stress-detecting smart helmet. Its early market is motorcyclists, cyclists, delivery riders, and ride-sharing operators who are concerned with the health of their riders. Its key sales point is real-time emotional safety in contrast to existing helmets, which only measure physical health. Its technology can be a premium add-on module in helmets or a pay-per-use subscription service on a mobile app with hardware. Partnerships with helmet manufacturers, insurance companies, and mobility platforms can create scalable use. It positions the solution as a safety product and a wellness service at the same time, with high entrepreneurial opportunity.
Comments					

System Overview



Appendix 1

Functional Requirements

- Continuous EEG signal acquisition during rides.
- BLE communication for seamless data transfer to the mobile app.
- Real-time stress classification using an ML model on EEG data.
- Instant notifications with suggestions (e.g., breaks, breathing cues, relaxation music).
- Live display of stress and emotional state in the mobile app UI.
- Configurable alerts (sound, text, vibration) for different user preferences.
- Logging of stress peaks for post-ride review and analysis.

Work Breakdown Structure

Phase	Weeks	Description
Literature & Analysis	1–2	Review EEG stress studies, finalize hardware and requirements.
Hardware Integration	3–5	Integrate EEG module, establish BLE communication with ESP32.
Model Development	6–9	Train and validate stress classification model with EEG datasets.
App Development	10–12	Implement mobile app UI for live monitoring and alerts.
Integration & Testing	13–15	Integrate hardware, model, and app; test stress detection on rides.
Evaluation & Refinement	16–18	Run user trials, gather feedback, finalize model and notifications.

Appendix 2

Key References

1. World Health Organization. *Global status report on road safety 2018*.
2. P. Casson, S. Yates, R. Smith, and J. Duncan, "Wearable EEG: What is it, why is it needed, and what does it entail?," *IEEE Signal Processing Magazine*, vol. 36, no. 1, pp. 38–47, 2019.
3. M. Hajinoroozi, J. M. Zhang and Y. Huang, "Driver's fatigue prediction by deep covariance learning from EEG," *2017 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Banff, AB, Canada, 2017, pp. 240-245, doi: 10.1109/SMC.2017.8122609.
4. R. K. Nath, H. Thapliyal, A. Caban-Holt and S. P. Mohanty, "Machine Learning Based Solutions for Real-Time Stress Monitoring," in *IEEE Consumer Electronics Magazine*, vol. 9, no. 5, pp. 34-41, 1 Sept. 2020, doi: 10.1109/MCE.2020.2993427.